

CS223 Computer Architecture & Organization

Cache Memory – Write Strategies



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Four cache memory design choices

- ❖ Where can a block be placed in the cache?
 - **Block Placement**
- ❖ How is a block found if it is in the upper level?
 - **Block Identification**
- ❖ Which block should be replaced on a miss?
 - **Block Replacement**
- ❖ What happens on a write?
 - **Write Strategy**

Block Placement

Block Number

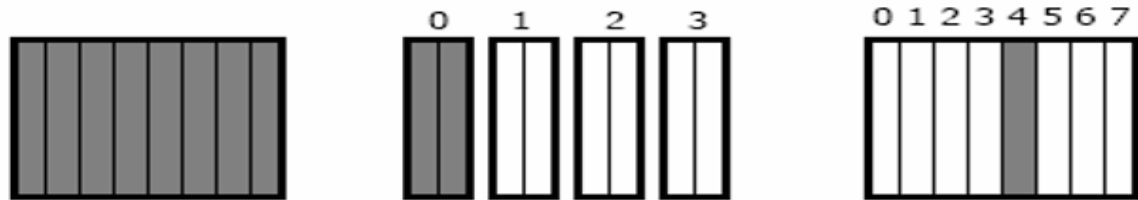
0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 2 2 2 2 2 2 2 2 2 2 2 3 3

Memory



Set Number

Cache



Fully
Associative

(2-way) Set
Associative

Direct
Mapped

block 12
can be placed

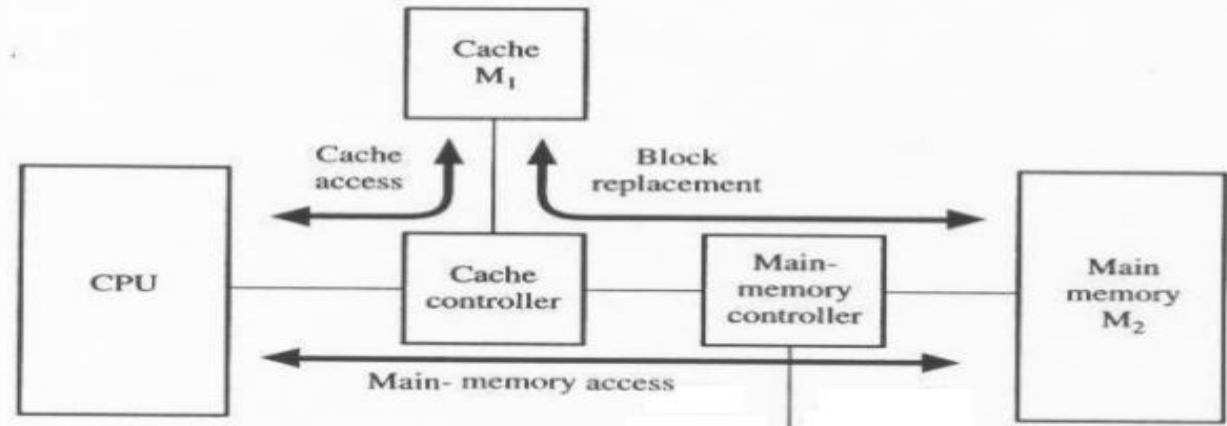
anywhere

anywhere in
set 0
($12 \bmod 4$)

only into
block 4
($12 \bmod 8$)

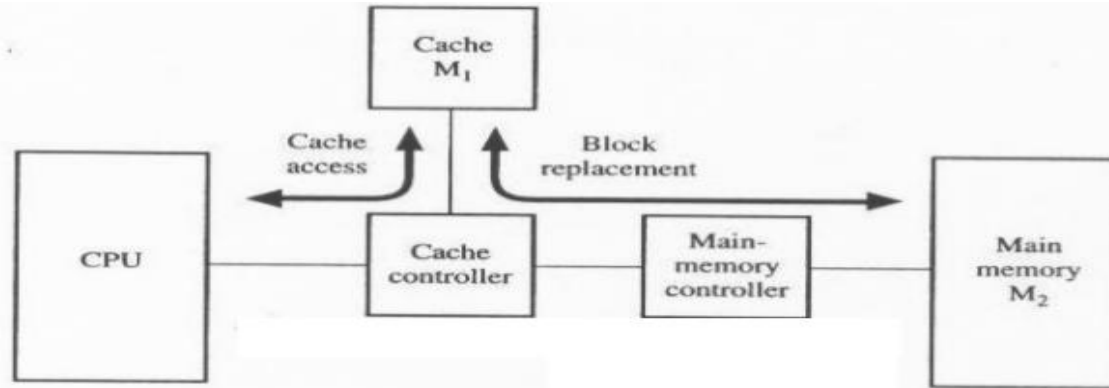
Look-aside vs Look through caches

- ❖ **Look-aside cache:** Request from processor goes to cache and main memory in parallel
- ❖ Cache and main memory both see the bus cycle
- ❖ On cache hit → processor loaded from cache, bus cycle terminates; On cache miss: processor & cache loaded from memory in parallel



Look-aside vs Look through caches

- ❖ **Look-through cache:** Cache checked first when processor requests data from memory
- ❖ On hit ☐ data loaded from cache: On miss ☐ cache loaded from memory, then processor loaded from cache

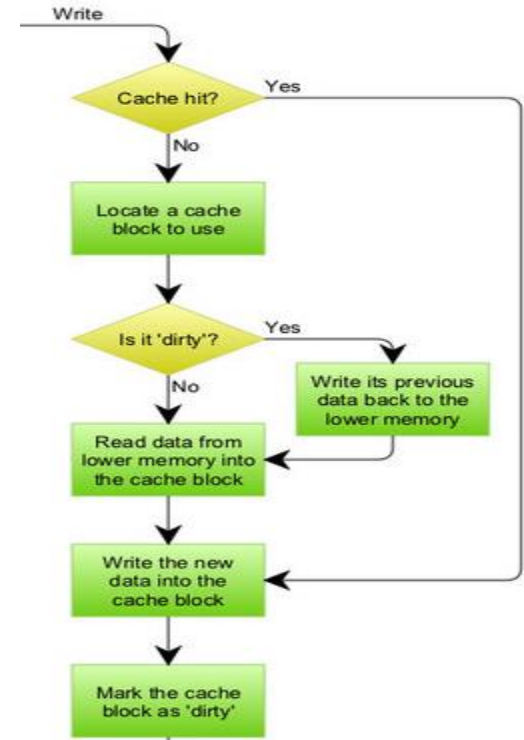
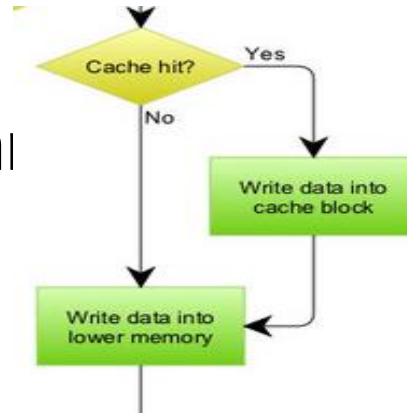


Write strategy

- ❖ **Write Hits → Write through vs Write back**
- ❖ **Write Miss → Write allocate vs No-Write allocate**
- ❖ **Write through:** The information is written to both the block in the cache and to the main memory
- ❖ Read misses do not need to write back evicted line contents
- ❖ **Write back:** The information is written only to the block in the cache. The modified cache block is written to main memory only when it is replaced.
- ❖ Have to maintain whether block clean or dirty. No extra work on repeated writes; only the latest value on eviction gets updated in main memory.

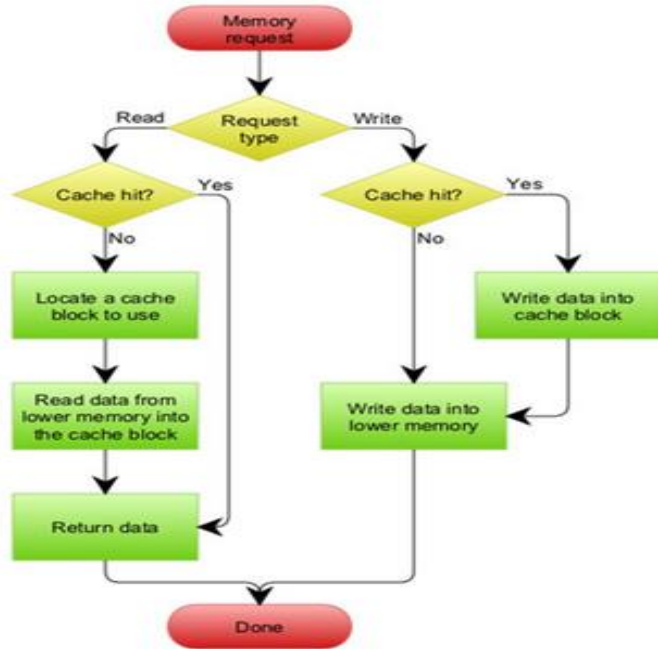
Write strategy

- ❖ **Write allocate:** The block is loaded into cache on a write miss.
- ❖ Used along with write back caches
- ❖ **No-Write allocate:** The block is modified in the main memory but not in cache
- ❖ Used along with write through caches

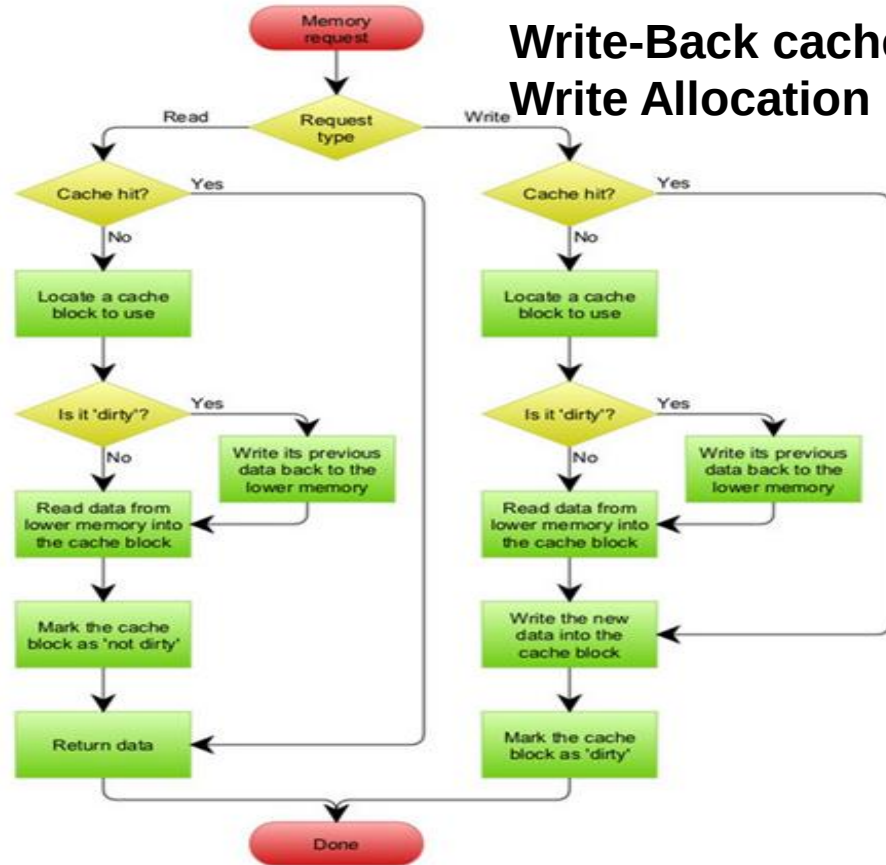


Write strategy

Write-Through cache with No-Write Allocation



Write-Back cache with Write Allocation



Types of Cache Misses

❖ Compulsory

- ❖ Very first access to a block
- ❖ Will occur even in an infinite cache

❖ Capacity

- ❖ If cache cannot contain all the blocks needed
- ❖ Misses in fully associative cache (due to the capacity)

❖ Conflict

- ❖ If too many blocks map to the same set
- ❖ Occurs in associative or direct mapped cache



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